

1. Diamond of stars

Write a program that reads an odd integer `n` and prints a centered diamond made of `*` with total height `n`.

Example (input 5):

```
*  
***  
*****  
***  
*
```

2. Hollow rectangle border

Read `rows` and `cols` and print a hollow rectangle using `*` for the border and spaces inside.

Example (input `rows=4 cols=7`):

```
*****  
*    *  
*    *  
*****
```

3. Right triangle + line sums

Read n and print a right triangle of numbers where line i contains i copies of i , then print | sum=<lineSum> at the end of each line.

Example (input 4):

```
1 | sum=1
22 | sum=4
333 | sum=9
4444 | sum=16
```

4. Odd-only multiplication table

Print the multiplication table for $\{1,3,5,7,9,11\} \times \{1,3,5,7,9,11\}$ aligned in a grid.

Example (first 3 rows/cols shown):

```
1 3 5
1 1 3 5
3 3 9 15
5 5 15 25
```

5. Snake-filled matrix

Read H and W, then print numbers from 1 to $H \times W$ in a “snake”

layout: row 0 left→right, row 1 right→left, alternating.

Example (input H=3 W=4):

```
1 2 3 4
8 7 6 5
9 10 11 12
```

6. Armstrong numbers in a range

Read integers L and R ($L \leq R$) and print all Armstrong numbers in [L, R], one per line.

Example (input 1 500):

```
1
2
3
4
5
6
7
8
9
153
370
371
```

407

7. Reverse and palindrome test (integer)

Read a positive integer x , compute its reverse using loops, then print the reverse and whether it is a palindrome.

Example (input 12321):

```
reverse=12321
```

```
PALINDROME
```

8. Prime listing and count

Read n and print all primes $\leq n$, then print
`count=<numberOfPrimes>`.

Example (input 20):

```
2 3 5 7 11 13 17 19
```

```
count=8
```